Gen AI Bootcamp day 2

🚀 Day 2 Homework Challenge – AI Product Intelligence System

Overview

Today, we built an AI-powered Product Intelligence System capable of:

- Understanding products from images
- Generating product metadata
- Finding similar products using embeddings and vector search

Your challenge is to extend this system by implementing **one or more advanced features**.

Dataset

https://www.kaggle.com/code/sahandakramipour/fashion-product-images-small

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Task 1: Smart Product Recommendation Engine

Problem Statement

Our current system finds products that are visually similar.

However, e-commerce platforms also recommend products that are commonly purchased together.

Build a recommendation system that suggests complementary products.

Example

Input Product:

```
Running Shoe
```

Output:

```
Recommended Products:
- Sports Socks
- Fitness Watch
- Water Bottle
```

Expected Deliverables

- Recommendation logic
- Explanation of how recommendations are generated
- Visualization of recommendations

Task 2: Unique Product Catalog Creation

Problem Statement

Large marketplaces often contain duplicate or near-duplicate products uploaded by different sellers.

Build a system that automatically creates a clean catalog containing only unique products.

Example

Input Dataset:

```
Blue Shirt A
Blue Shirt B
Blue Shirt C
Running Shoe A
Running Shoe B
```

Output Catalog:

```
Blue Shirt
Running Shoe
```

Expected Deliverables

- Identify duplicate products using embeddings
- Group similar products together
- Generate a final catalog of unique products

Task 3: Reverse Product Search

Problem Statement

Instead of searching using an image, allow users to search using text.

Example

Input Query:

```
blue casual shirt
```

Output:

```
Top Matching Products:
1. Men's Blue Casual Shirt
2. Blue Checked Shirt
3. Slim Fit Blue Shirt
```

Expected Deliverables

- Text-based search interface
- Retrieval of relevant products
- Display of matching products

Hint : Explore CLIP Image and Text Embeddings

Submission Requirements

Submit a zip file containing the following:

1. Source Code / Kaggle Notebook (.ipynb file)
2. Report (Short explanation of all tasks and screenshots of results)

Judging Criteria

Criteria	Percentage
Correctness	40
Creativity	10
Technical Implementation	20
Presentation & Explanation	30
